



نظم التشغيل

12:2

الخميس 24/6/2021

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Faculty of Computers & Information, Assiut University

3rd Level

Final Exam

Duration: 2 hours

* Required

* This form will record your name, please fill your name.

1

* الإسم الرباعي (بالعربي فقط)

2

* رقم الجلوس

3

* المستوى

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ رابعة 2013
- ☐ رابعة 2014
- ☐ رابعة 2015
- ☐ رابعة 2016
- ☐ رابعة 2017

4

* البرنامج

- ☐ عام
- ☐ بايو
- ☐ هندسة

* رقم المعمل

☐ ج٠

☐ د٠

☐ اب

☐ اد

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☐ د٢

☐ ه٢

☐ أ٣

☐ ب٣

☐ ج٣

☐ د٣

☐ ه٣

☐ أ٤

☐ ب٤

6

* رقم الكمبيوتر

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

8

In Operating Systems, which of the following is/are CPU scheduling algorithms?
(2 Points)

- ☐ Round Robin
- ☐ Shortest Job First
- ☐ Priority
- ☐ All of the mentioned

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In operating system, each process has its own _____
(2 Points)

- ☐ address space and global variables
- ☐ open files
- ☐ pending alarms, signals, and signal handlers
- ☐ all of the mentioned

10

What is interprocess communication?
(2 Points)

- ☐ communication within the process
- ☐ communication between two processes
- ☐ communication between two threads of same process
- ☐ none of the mentioned

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A set of processes is deadlock if _____
(2 Points)

- ☐ each process is blocked and will remain so forever
- ☐ each process is terminated
- ☐ all processes are trying to kill each other
- ☐ none of the mentioned

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The number of processes completed per unit time is known as _____
(2 Points)

- ☐ Capacity
- ☐ Throughput
- ☐ Output
- ☐ Efficiency

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What is the degree of multiprogramming?
(2 Points)

- ☐ the number of processes in the ready queue
- ☐ the number of processes in the I/O queue
- ☐ the number of processes executed per unit time
- ☐ the number of processes in memory

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When the process issues an I/O request _____
(2 Points)

- ☐ It is placed in an I/O queue
- ☐ It is placed in a waiting queue
- ☐ It is placed in the Job queue
- ☐ It is placed in the ready queue

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What is a long-term scheduler?

(2 Points)

- ☐ It selects processes which must be brought into the ready queue
- ☐ It selects processes which must be executed next and allocates CPU
- ☐ It selects processes which must be removed from memory by swapping
- ☐ None of the mentioned

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What is a short-term scheduler?

(2 Points)

- ☐ It selects which process must be brought into the ready queue
- ☐ It selects which process must be executed next and allocates CPU
- ☐ It selects which process must be removed from memory by swapping
- ☐ None of the mentioned

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Suppose that a process is in "Blocked" state waiting for some I/O service. When the service is completed, it goes to the _____

(2 Points)

- ☐ Suspended state
- ☐ Terminated state
- ☐ Running state
- ☐ Ready state

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When several processes access the same data concurrently and the outcome of the execution depends on the order in which the access takes place is called _____
(2 Points)

- ☐ essential condition
- ☐ dynamic condition
- ☐ race condition
- ☐ critical condition

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If a process is executing in its critical section, then no other processes can be executing in their critical section. What is this condition called?
(2 Points)

- ☐ mutual exclusion
- ☐ asynchronous exclusion
- ☐ synchronous exclusion
- ☐ critical exclusion

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Which one of the following is a synchronization tool?

(2 Points)

- ☒ semaphore
- ☐ thread
- ☐ pipe
- ☐ socket

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DMA is used for _____

(2 Points)

- ☒ High speed devices
- ☐ Low speed devices
- ☐ Utilizing CPU cycles
- ☐ All of the mentioned

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The interval from the time of submission of a process to the time of completion is termed as _____

(2 Points)

- ☐ waiting time
- ☒ turnaround time
- ☐ response time
- ☐ throughput

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Which algorithm is defined in Time quantum?

(2 Points)

- ☒ round robin scheduling algorithm
- ☐ priority scheduling algorithm
- ☐ multilevel queue scheduling algorithm
- ☐ shortest job scheduling algorithm

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A process is selected from the _____ queue by the _____ scheduler, to be executed

(2 Points)

- ☐ ready, long term
- ☒ ready, short term
- ☐ blocked, short term
- ☐ wait, long term

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What is Response time?

(2 Points)

- ☐ the total time taken from the submission time till the completion time
- ☒ the total time taken from the submission time till the first response is produced
- ☐ the total time taken from submission time till the response is output
- ☐ none of the mentioned

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Round robin scheduling falls under the category of _____
(2 Points)

- ☐ Non-preemptive scheduling
- ☒ Preemptive scheduling
- ☐ All of the mentioned
- ☐ None of the mentioned

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The real difficulty with SJF in short term scheduling is _____
(2 Points)

- ☐ it is too good an algorithm
- ☒ knowing the length of the next CPU request
- ☐ it is too complex to understand
- ☐ none of the mentioned

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What is 'Aging'?
(2 Points)

- ☐ keeping track of what pages are currently residing in memory
- ☒ increasing the priority of jobs to ensure termination in a finite time
- ☐ keeping track of cache contents
- ☐ keeping track of how many times a given page is referenced

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Which of the following scheduling algorithms gives minimum average waiting time?
(2 Points)

- ☒ SJF
- ☐ FCFS
- ☐ Round – robin
- ☐ Priority

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Which of the following conditions must be satisfied to solve the critical section problem?
(2 Points)

- ☐ Mutual Exclusion
- ☐ Progress
- ☐ Bounded Waiting
- ☒ All of the mentioned

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A binary semaphore is a semaphore with integer values _____
(2 Points)

- ☐ -2
- ☐ 2
- ☐ -1
- ☒ 1

32

A system is in a safe state only if there exists a _____
(2 Points)

- ☐ safe allocation
- ☐ safe resource
- ☐ safe sequence
- ☐ all of the mentioned

33

If no cycle exists in the resource allocation graph _____
(2 Points)

- ☐ then the system will not be in a safe state
- ☐ then the system will be in a safe state
- ☐ all of the mentioned
- ☐ none of the mentioned

34

The wait-for graph is a deadlock detection algorithm that is applicable when _____
(2 Points)

- ☐ all resources have a single instance
- ☐ all resources have multiple instances
- ☐ all resources have a single 7 multiple instances
- ☐ all of the mentioned

35

The address generated by the CPU is referred to as _____
(2 Points)

- ☐ Physical address
- ☒ Logical address
- ☐ Neither physical nor logical
- ☐ None of the mentioned

36

The run time mapping from virtual to physical addresses is done by a hardware device called the _____
(2 Points)

- ☐ Virtual to physical mapper
- ☒ Memory management unit
- ☐ Memory mapping unit
- ☐ None of the mentioned

37

Program always deals with _____
(2 Points)

- ☐ absolute address
- ☐ physical address
- ☒ logical address
- ☐ relative address

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What is compaction?
(2 Points)

- ☐ a paging technique
- ☐ a technique for overcoming fatal error
- ☐ a technique for overcoming internal fragmentation
- ☐ a technique for overcoming external fragmentation

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In contiguous memory allocation _____
(2 Points)

- ☐ each process is contained in a single contiguous section of memory
- ☐ all processes are contained in a single contiguous section of memory
- ☐ the memory space is contiguous
- ☐ none of the mentioned

40

The first fit, best fit and worst fit are strategies to select a _____
(2 Points)

- ☐ process from a queue to put in memory
- ☐ processor to run the next process
- ☐ free hole from a set of available holes
- ☐ all of the mentioned

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In internal fragmentation, memory is internal to a partition and _____
(2 Points)

- ☐ is being used
- ☒ is not being used
- ☐ is always used
- ☐ none of the mentioned

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_____ is generally faster than _____ and _____
(2 Points)

- ☒ first fit, best fit, worst fit
- ☐ best fit, first fit, worst fit
- ☐ worst fit, best fit, first fit
- ☐ none of the mentioned

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Every address generated by the CPU is divided into two parts. They are _____
(2 Points)

- ☐ frame bit & page number
- ☒ page number & page offset
- ☐ page offset & frame bit
- ☐ frame offset & page offset

44

The offset 'd' of the logical address must be _____
(2 Points)

- ☐ between 0 and segment limit
- ☐ greater than the segment number
- ☐ greater than segment limit
- ☐ between 0 and the segment number

45

When high priority task is indirectly preempted by medium priority task effectively inverting the relative priority of the two tasks, the scenario is called _____
(2 Points)

- ☐ priority exchange
- ☐ priority inversion
- ☐ priority removal
- ☐ priority modification

Which of the following scheduling policy is well suited for a time-shared operating system?

(2 Points)

- ☒ Round robin
- ☐ Elevator
- ☐ Shortest job first
- ☐ FCFS

Assuming the above process being scheduled with the SJF scheduling algorithm (non-preemptive). The waiting time for process C is:
(2 Points)

Process	Arrival Time	Processing Time
A	0	4
B	1	5
C	2	4
D	3	2

- ☐ 2
- ☒ 4 correct
- ☐ 6
- ☐ 8
- ☐ None of the mentioned

Assume Quantum value two, given a RR scheduler, what is the response time of job B?

(2 Points)

Process	Arrival Time	Processing Time
A	0	4
B	1	5
C	2	4
D	3	2

☐ 0

☒ 1 correct

☐ 2

☐ 3

☐ None of the mentioned

Given a FIFO scheduler, what is the average response time of the four jobs?
(2 Points)

Process	Arrival Time	Processing Time
A	0	4
B	1	5
C	2	4
D	3	2

- ☐ 4.5
- ☒ 5 correct
- ☐ 6.5
- ☐ 8
- ☐ None of the mentioned

50

Virtual memory is
(2 Points)

- ☐ a type of memory used in super computers
- ☐ an extremely large main memory
- ☐ an illusion of an extremely large memory
- ☐ an extremely large secondary memory

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Necessary conditions for deadlock are
(2 Points)

- ☐ non-preemption and circular wait
- ☐ Mutual exclusion
- ☐ both (a) and (b)
- ☐ none of the above

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The following system of four processes with two resources:

If the availability vector is [2 3], is the system above deadlocked?

(2 Points)

Current allocation matrix:	Current request matrix:
P ₁ 1 3	P ₁ 1 2
P ₂ 4 1	P ₂ 4 3
P ₃ 1 2	P ₃ 1 7
P ₄ 2 0	P ₄ 5 1

☐ Yes

☐ No

53

The following system of four processes with two resources:

If the availability vector is [2 5], is the system above deadlocked?

(2 Points)

Current allocation matrix:	Current request matrix:
P ₁ 1 3	P ₁ 1 2
P ₂ 4 1	P ₂ 4 3
P ₃ 1 2	P ₃ 1 7
P ₄ 2 0	P ₄ 5 1

☐ Yes

☐ No

54

A small computer has 3 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,7,6,3,2,1,2,3,6. How many page faults using FIFO algorithm? (2 Points)

- ☐ 8
- ☐ 10
- ☒ 12
- ☐ 14
- ☐ None of the mentioned

55

A small computer has 3 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,7,6,3,2,1,2,3,6. How many page faults using least-recently-used (LRU) algorithm? (2 Points)

- ☐ 8
- ☐ 10
- ☒ 12
- ☐ 14
- ☐ None of the mentioned

56

A small computer has 3 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,7,6,3,2,1,2,3,6. How many page faults using optimal page replacement algorithm?

(2 Points)

☐ 8

☒ 10

☐ 12

☐ 14

☐ None of the mentioned

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If a physical address is 32 bits and each page is 32KB, the top bits exactly designate the physical page number

(2 Points)

☐ 10

☒ 17

☐ 19

☐ 32